# Three-step equations

**1** Solve the following equations:

**a** $3(4s+1)=-21$

**b** $3(2x+1)=15$

**c** $2(3t-7)=16$

**d** $-5(3x+8)=-10$

**e** $2(-3g+7)=-16$

**f** $-2(3x-5)=28$

**g** $-3(-2h-5)=33$

**h** $-3(-2x+7)=-45$

**i** $4(5x-1)=-24$

**j** $-4(5x+6)=-104$

**k** $2(7+2x)=38$

**l** $-3(4-x)=-12$

**m** $5(2x-6)=100$

**n** $6(3x+15)=-72$

**o** $9(8-x)=45$

**p** $10(7-3x)=20$

**q** $20=2(3x+11)$

**r** $16=4(2x+9)$

**s** $18=3(2x+14)$

**t** $-5=5(3x+1)$

**u** $26=13(8-4x)$

**v** $32=8(16-5x)$

**2** Ryan attempted to solve the equation $9(4x-6)=18$. His working is shown below:

$$\begin{aligned} 9(4x-6)&=18\\ 4x-6&=9\\ 4x&=15\\ x&=\frac{15}{4} \end{aligned}$$

**a** What was his mistake?

**b** Solve the equation correctly.

**3** Solve the following equations:

**a** $3(w+8)+5=44$

**b** $9(x-7)+8=26$

**c** $-5(g+4)+9=-41$

**d** $-7(x+5)+4=-17$

**e** $6(q+1)-5=-23$

**f** $4(x+6)-8=48$

**g** $-8(x+4)-3=-59$

**h** $-2(x-2)-7=-29$

**i** $2(x-9)-4=16$

**j** $5(x-3)-7=-52$

**4** Solve the following equations:

**a** $-\frac{y}{3}+10=17$

**b** $-\frac{u}{4}+15=8$

**c** $\frac{5x}{8}-9=-4$

**d** $\frac{8c}{3}+5=-11$

**e** $\frac{5x}{3}+11=21$

**f** $\frac{-3c}{4}+5=14$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

g $-\frac{3x}{4}+5=-7$

h $\frac{3x}{4}-5=-11$

i $\frac{4x}{7}-6=-14$

j $-\frac{5x}{6}-6=14$

**5** Solve the following equations:

a $\frac{x-9}{5}+4=7$

b $\frac{x-3}{5}-8=-7$

c $\frac{x+2}{3}-25=-22$

d $\frac{x+16}{3}+5=3$

e $\frac{2x-12}{3}=0$

f $\frac{3x+6}{2}=15$

g $\frac{8x+4}{5}=-4$

h $\frac{-3t-6}{7}=-3$

i $\frac{-13-4r}{3}=-15$

j $\frac{-9+5x}{2}=-22$

**6** Solve the following equations:

a $\frac{2(x-4)}{3}=8$

b $\frac{5(x+1)}{2}=10$

c $\frac{-3(x-3)}{4}=0$

d $\frac{6(x+5)}{7}=12$

e $\frac{9(x+10)}{2}=-6$

f $\frac{4(x-2)}{3}=-3$

g $\frac{7(x-9)}{5}=14$

h $\frac{2(x+11)}{9}=8$

**7** Solve the following equations:

a $2\left(\frac{x}{3}+1\right)=6$

b $5\left(\frac{x}{2}+7\right)=60$

c $4\left(\frac{x}{5}-3\right)=-12$

d $3\left(\frac{x}{4}-10\right)=15$

e $18=2\left(\frac{x}{5}-7\right)$

f $40=10\left(\frac{x}{6}-1\right)$

g $8\left(4+\frac{x}{2}\right)=48$

h $-4\left(12-\frac{x}{3}\right)=-64$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647